

CodeAlpha IoT Internship – Final Submission Report

- **Student Information**

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Domain: Internet of Things (IoT)

Organization: CodeAlpha Internship Program

INTERNET OF THINGS (IoT) – INTERNSHIP REPORT

- **TASK 1: IoT in Real Life Smart Agriculture**

Introduction

The Internet of Things (IoT) is a technology that connects physical devices to the internet to collect, exchange, and analyze data. It enables automation and smart decision-making across various industries, especially agriculture.

Smart Agriculture System

Smart Agriculture integrates sensors, microcontrollers, and internet connectivity to improve farming efficiency and productivity.

Smart Irrigation System

- Soil Moisture Sensor measures soil condition
- Arduino processes sensor data
- If soil is dry → Water Pump ON
- If soil is wet → Water Pump OFF

Result: Water conservation and improved crop yield

Crop Monitoring System

IoT sensors monitor temperature, humidity, and soil conditions in real time.

Farmers can access this data via mobile applications.

Benefits:

- Improved decision-making
- Efficient fertilizer usage
- Better pest control

Livestock Monitoring

IoT wearable devices help in:

- Tracking animal location
- Monitoring health conditions
- Sending emergency alerts

Advantages

- Water saving
- Increased productivity
- Reduced labor cost
- Smart farming decisions

Challenges

- High initial cost
- Internet connectivity issues in rural areas
- Technical expertise requirement

Conclusion

IoT significantly enhances agricultural efficiency and plays a vital role in modern smart farming systems.

TASK 2: Sensor-Based Simulation Automatic Light Control System

Components

- LDR Sensor
 - Arduino Uno
 - LED
 - Resistor
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Working Principle

Light intensity is detected using an LDR sensor.

- Low light → LED ON
 - High light → LED OFF
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Arduino Code

```
int ldrPin = A0;
int ledPin = 9;
int threshold = 500;

void setup() {
  pinMode(ledPin, OUTPUT);
}

void loop() {
  int ldrValue = analogRead(ldrPin);

  if (ldrValue < threshold) {
    digitalWrite(ledPin, HIGH);
```

```
    }  
    else {  
        digitalWrite(ledPin, LOW);  
    }  
  
    delay(500);  
}
```

Conclusion

This system demonstrates basic IoT automation and energy-saving applications.

TASK 3: IoT Prototype – Smart Door Lock System

Components

- Arduino Uno
 - Keypad
 - Servo Motor
 - Buzzer
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Working Principle

- User enters password
 - Arduino verifies input
 - Correct password → Door Unlock
 - Wrong password → Alarm ON
-

Arduino Code

```
#include <Servo.h>  
  
Servo doorServo;  
  
int buzzer = 8;  
String password = "1234";  
String input = "";
```

```
void setup() {  
  doorServo.attach(9);  
  pinMode(buzzer, OUTPUT);  
}  
  
void loop() {  
  if (Serial.available()) {  
    char c = Serial.read();  
    input += c;  
  
    if (input.length() == 4) {  
      if (input == password) {  
        doorServo.write(90);  
        digitalWrite(buzzer, LOW);  
      }  
      else {  
        digitalWrite(buzzer, HIGH);  
      }  
  
      delay(3000);  
      doorServo.write(0);  
      input = "";  
    }  
  }  
}
```

Advantages

- Improved security
 - Automated access control
 - Smart home integration
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Future Improvements

- Mobile app integration (ESP8266)
 - Cloud-based security system
 - Face recognition upgrade
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TASK 4: IoT Case Study – Security Challenges

Introduction

IoT devices are connected to the internet, making security a critical aspect of system design.

Security Issues

- Weak default passwords
 - Data privacy risks
 - Lack of software updates
 - Network attacks
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Solutions

- Strong password policies
 - Data encryption
 - Regular firmware updates
 - Secure cloud systems
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Future of IoT Security

Artificial Intelligence and Machine Learning will enhance real-time threat detection and prevention.

Conclusion

Security is essential for safe and reliable IoT systems.

SUBMISSION DETAILS

Folder Name:

CodeAlpha_Muhammad_Abdullah_IoT

Includes:

- Final PDF Report
- Arduino Code Files
- Screenshots / Simulation Evidence
- Case Study Document

Sharing Setting:

Anyone with the link can view